Homework 1

Solutions

1. (Dunning-Kruger effect) Since the means of all variables are zero,

$$\begin{aligned}\mathrm{Cov}\left[\tilde{t},\tilde{d}\right] &= \mathrm{E}\left[\tilde{t}\left(\tilde{s}-\tilde{t}\right)\right] && (1)\\ &= \mathrm{E}\left[\tilde{t},\tilde{s}\right]-\mathrm{E}\left[\tilde{t}^2\right] && (2)\\ &= -\mathrm{Var}\left[\tilde{t}\right]=1, && (3)\end{aligned}$$

and by the uncorrelation assumption

$$\begin{aligned}\mathrm{Var}\left[\tilde{d}\right] &= \mathrm{Var}\left[\tilde{s}-\tilde{t}\right] && (4)\\ &= \mathrm{Var}\left[\tilde{s}\right]+\mathrm{Var}\left[-\tilde{t}\right] && (5)\\ &= \mathrm{Var}\left[\tilde{s}\right]+\mathrm{Var}\left[\tilde{t}\right] && (6)\\ &= 2. && (7)\end{aligned}$$

Consequently,

$$\begin{aligned}\rho_{\tilde{t},\tilde{d}} &= \frac{\mathrm{Cov}\left[\tilde{t},\tilde{d}\right]}{\sqrt{\mathrm{Var}\left[\tilde{t}\right]\mathrm{Var}\left[\tilde{d}\right]}} && (8)\\ &= \frac{-1}{\sqrt{2}}=-0.707. && (9)\end{aligned}$$

The study finds a strong negative correlation, even though the true and self-evaluated competences are uncorrelated!

2. (Three variables)

   (a) Let $\tilde{a}$, $\tilde{b}$ and $\tilde{c}$ be random variables with zero mean and unit variance, which are pairwise uncorrelated. We define:

$$\begin{aligned}\tilde{w} &:= \tilde{a}+\alpha\tilde{b}, && (10)\\ \tilde{y} &:= \tilde{b}+\beta\tilde{c}, && (11)\\ \tilde{z} &:= \gamma\tilde{a}+\tilde{c}, && (12)\end{aligned}$$

   where α, β and γ are real-valued constants. By linearity of expectation

$$\begin{aligned}\mathrm{E}\left[\tilde{w}\right] &= \mathrm{E}\left[\tilde{a}\right]+\alpha\mathrm{E}[\tilde{b}]=0, && (13)\\ \mathrm{E}\left[\tilde{y}\right] &= \mathrm{E}[\tilde{b}]+\beta\mathrm{E}\left[\tilde{c}\right]=0, && (14)\\ \mathrm{E}\left[\tilde{z}\right] &= \gamma\mathrm{E}\left[\tilde{a}\right]+\mathrm{E}\left[\tilde{c}\right]=0. && (15)\end{aligned}$$

   Since $\tilde{a}$, $\tilde{b}$ and $\tilde{c}$ have zero mean and are pairwise uncorrelated,

$$\mathrm{E}[\tilde{a}\tilde{b}]=\mathrm{E}\left[\tilde{a}\tilde{c}\right]=\mathrm{E}[\tilde{b}\tilde{c}]=0. \quad (16)$$

Consequently, again by linearity of expectation,

$$\begin{aligned}
\mathrm{E}\left[\tilde{w}\tilde{y}\right] &= \mathrm{E}[\tilde{a}\tilde{b}] + \beta\mathrm{E}\left[\tilde{a}\tilde{c}\right] + \alpha\mathrm{E}[\tilde{b}^2] + \alpha\beta\mathrm{E}[\tilde{b}\tilde{c}] && (17)\\
&= \alpha, && (18)\\
\mathrm{E}\left[\tilde{w}\tilde{z}\right] &= \gamma\mathrm{E}\left[\tilde{a}^2\right] + \mathrm{E}\left[\tilde{a}\tilde{c}\right] + \gamma\alpha\mathrm{E}[\tilde{a}\tilde{b}] + \alpha\mathrm{E}[\tilde{b}\tilde{c}] && (19)\\
&= \gamma, && (20)\\
\mathrm{E}\left[\tilde{y}\tilde{z}\right] &= \gamma\mathrm{E}[\tilde{a}\tilde{b}] + \mathrm{E}[\tilde{b}\tilde{c}] + \gamma\beta\mathrm{E}\left[\tilde{a}\tilde{c}\right] + \beta\mathrm{E}\left[\tilde{c}^2\right] && (21)\\
&= \beta. && (22)
\end{aligned}$$

If we choose α and β to be positive and γ negative, then $\tilde{w}$ and $\tilde{y}$ are positively correlated, and so are $\tilde{y}$ and $\tilde{z}$, but $\tilde{w}$ and $\tilde{y}$ are negatively correlated.

(b)

Some sample data:

$$\begin{bmatrix} 1.5 & -1.8 & -3 \\ 0.6 & 0 & -0.6 \\ -1 & -2 & -2 \end{bmatrix}$$

Sample correlation matrix:

$$\begin{pmatrix} 1.0000000 & 0.2485479 & -0.2642224 \\ 0.2485479 & 1.0000000 & 0.8685247 \\ -0.2642224 & 0.8685247 & 1.0000000 \end{pmatrix}$$

3. (Averaging noisy data)

   (a) Let $\tilde{w} := \sum_{i=1}^{n} \tilde{y}_i = n\tilde{x} + \sum_{i=1}^{n} \tilde{z}_i$. We need to find the linear MMSE estimator of $\tilde{x}$ given $\tilde{w}$. We have

$$\begin{aligned}
\mathrm{Var}[\tilde{w}] &= \mathrm{Var}\left[n\tilde{x} + \sum_{i=1}^{n} \tilde{z}_i\right] && (23)\\
&= n^2 + n\sigma^2, && (24)\\
\mathrm{E}[\tilde{x}\tilde{w}] &= \mathrm{E}[n\tilde{x}^2] + \sum_{i=1}^{n} \mathrm{E}[\tilde{x}\tilde{z}_i] && (25)\\
&= n, && (26)\\
\rho_{\tilde{x},\tilde{w}} &= \frac{\mathrm{E}[\tilde{x}\tilde{w}]}{\sigma_{\tilde{x}}\sigma_{\tilde{w}}} && (27)\\
&= \frac{n}{\sqrt{n^2 + n\sigma^2}} && (28)\\
&= \sqrt{\frac{n}{n + \sigma^2}}. && (29)
\end{aligned}$$

The linear MMSE estimate of $\tilde{x}$ given $\tilde{w} = w$ equals

$$\ell_{\text{MMSE}}(w) = \frac{\rho_{\tilde{x},\tilde{w}}\sigma_{\tilde{x}}w}{\sigma_{\tilde{w}}} \tag{30}$$

$$= \frac{w}{n+\sigma^2}. \tag{31}$$

Therefore the optimal α equals $\frac{1}{n+\sigma^2}$.

(b) As $\sigma^2 \to 0$, the estimator is just the average of the measurements. As $\sigma^2 \to \infty$ the estimator tends to zero: the measurements are ignored because they are just noise.

(c) The MSE equals

$$\sigma_{\tilde{x}}^2(1-\rho_{\tilde{x},\tilde{w}}^2) = 1 - \frac{n}{n+\sigma^2} \tag{32}$$

$$= \frac{\sigma^2}{n+\sigma^2}, \tag{33}$$

which tends to zero as $n \to \infty$.

4. (Interference)

(a) Since $\tilde{w}$ and $\tilde{a}$ are independent

$$\mathrm{E}\left[\tilde{y}\right] = \mathrm{E}\left[\tilde{w}\tilde{a}\right] \tag{34}$$

$$= \mathrm{E}\left[\tilde{w}\right]\mathrm{E}\left[\tilde{a}\right] \tag{35}$$

$$= 0 \tag{36}$$

because $\mathrm{E}\left[\tilde{w}\right] = -0.5 + 0.5 = 0$. Similarly,

$$\mathrm{E}\left[\tilde{a}\tilde{y}\right] = \mathrm{E}\left[\tilde{w}\tilde{a}^2\right] \tag{37}$$

$$= \mathrm{E}\left[\tilde{w}\right]\mathrm{E}\left[\tilde{a}^2\right] \tag{38}$$

$$= 0. \tag{39}$$

Therefore,

$$\mathrm{Cov}\left[\tilde{a},\tilde{y}\right] = \mathrm{E}\left[\tilde{a}\tilde{y}\right] - \mathrm{E}\left[\tilde{a}\right]\mathrm{E}\left[\tilde{y}\right] \tag{40}$$

$$= 0. \tag{41}$$

As a result, the linear MMSE estimator of $\tilde{a}$ given $\tilde{y} = y$ is

$$\ell_{\text{MMSE}}(y) = \mathrm{E}\left[\tilde{a}\right] \tag{42}$$

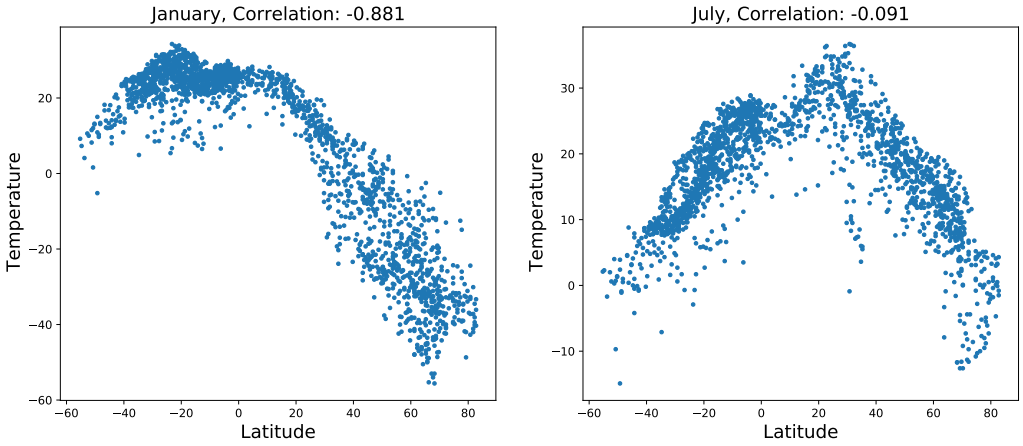
$$= \mu. \tag{43}$$

(b) The MSE equals

$$\text{MSE} := \mathrm{E}\left[\left(\tilde{a} - \ell_{\text{MMSE}}(\tilde{y})\right)^2\right] \tag{44}$$

$$= \mathrm{E}\left[\left(\tilde{a}-\mu\right)^2\right] \tag{45}$$

$$= \sigma^2. \tag{46}$$

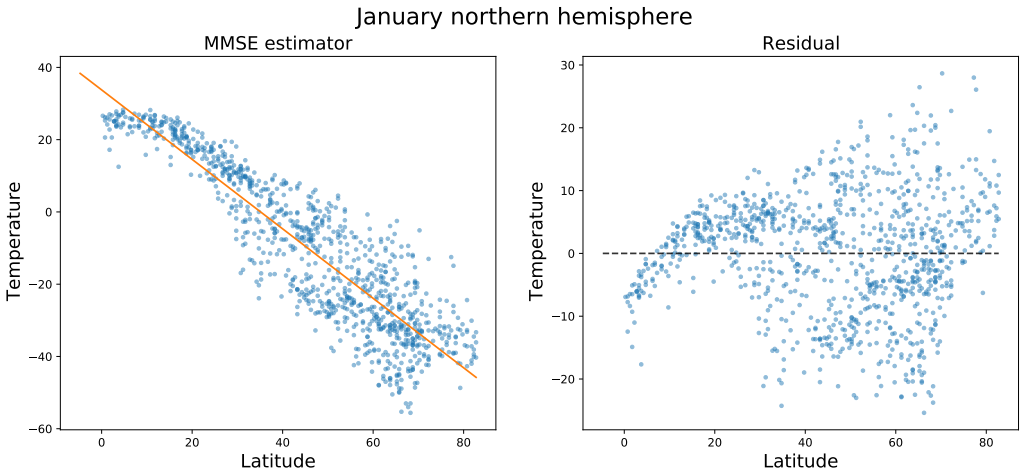


(c) Consider the nonlinear estimator $|\tilde{y}|$. Since $\tilde{a}$ is nonnegative, $|y| = \tilde{a}$, so the MSE is zero.

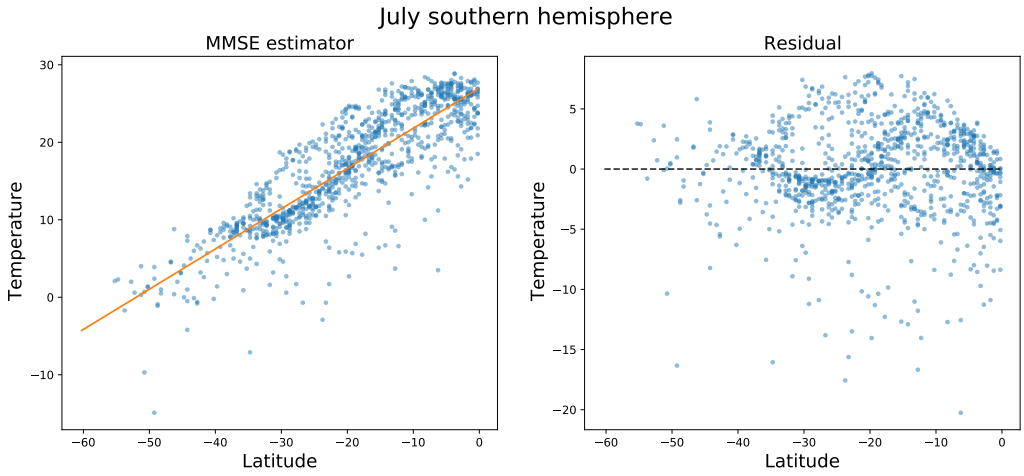
5. (Temperature)

(a)

(b) The correlation is 0.85.



(c) The correlation is -0.91.


July southern hemisphere
MMSE estimator
Residual
Temperature
Latitude
30
20
10
0
−10
5
−5
−15
−20
−60
−50
−40
−30
−20
−10
0